A Consistent Forward–Adjoint Orbital Metric

Implementation note

16 October 2026

1 Physical tangent

Let A and B contain the inactive and active raw orbital coefficients in the atomic-orbital (AO) basis, and let S be the AO overlap matrix. All quantities without a variation symbol are fixed at the accepted point. Define

$$H_i = A^\mathsf{T} S A = R_i^\mathsf{T} R_i, \qquad S = L^\mathsf{T} L, \tag{1}$$

$$P = I - A H_i^{-1} A^\mathsf{T} S, \qquad D = PB, \qquad K = H_i^{-1} A^\mathsf{T} S B. \tag{2}$$

Here R_i and L are upper-triangular Cholesky factors, P is the inactive complement, and D contains the projected active orbitals. Given allowed raw-coefficient variations $(\delta A, \delta B)$, the physical derivatives are

$$h = P\delta A, \tag{3}$$

$$v = P\delta B - hK - A H_i^{-1} h^\mathsf{T} S B. \tag{4}$$

The last term in Eq. (4) is part of the changing inactive projector; it is not an optional structure response.

For an individual active ray $d_j = D[:, j]$, write

$$n_j = d_j^\mathsf{T} S d_j, \qquad Q_j = I - d_j d_j^\mathsf{T} S / n_j. \tag{5}$$

The tangent feature map $\mathcal{J}$ is

$$F_i = L h R_i^{-1}, \qquad f_{a,j} = L Q_j v_j / \sqrt{n_j}, \qquad \mathcal{J}(\delta A, \delta B) = (F_i, F_a). \tag{6}$$

It measures inactive-subspace motion and normalized active-ray motion. This ray representation applies to selected valence-bond structure spaces; it assumes no closure under active-orbital transformations.

Only when the orbital manifold has the complete full-AO active-space gauge may the active rays be replaced by their span. In that case define

$$H_a = D^\mathsf{T} S D = R_a^\mathsf{T} R_a, \qquad Q_a = I - D H_a^{-1} D^\mathsf{T} S, \qquad F_a = L Q_a v R_a^{-1}. \tag{7}$$

Completing a computational spin-string carrier does not establish this physical gauge for an arbitrarily selected structure space.

2 Strict adjoint

Let Z_i and Z_a be feature covectors. Reverse precisely the operations in Eqs. (6)–(7):

$$W_i = L^\mathsf{T} Z_i R_i^{-\mathsf{T}}, \tag{8}$$

$$w_{a,j} = Q_j^\mathsf{T} L^\mathsf{T} z_{a,j} / \sqrt{n_j} \quad \text{(individual rays)}, \tag{9}$$

$$W_a = Q_a^\mathsf{T} L^\mathsf{T} Z_a R_a^{-\mathsf{T}} \quad \text{(complete active span)}. \tag{10}$$

The raw-coefficient covectors are then

$$\overline{B} = P^{\mathsf{T}} W_a, \tag{11}$$

$$\overline{A} = P^{\mathsf{T}} \left[W_i - W_a K^{\mathsf{T}} - SBW_a^{\mathsf{T}} (AH_i^{-1}) \right]. \tag{12}$$

These equations follow by the Frobenius identity

$$\langle Z_i, F_i \rangle_{\mathrm{F}} + \langle Z_a, F_a \rangle_{\mathrm{F}} = \langle \overline{A}, \delta A \rangle_{\mathrm{F}} + \langle \overline{B}, \delta B \rangle_{\mathrm{F}}. \tag{13}$$

In particular, the last term in Eq. (12) uses the transpose of the *stored forward factor* AH_i^{-1}. It must not independently reconstruct that transpose by assuming an explicitly computed inverse is exactly symmetric in floating-point arithmetic.

3 Metric action and ownership

If U denotes the existing nonredundant sparse lift, the reduced metric is

$$G = U^{\mathsf{T}} \mathcal{J}^* \mathcal{J} U, \qquad p^{\mathsf{T}} G p = \|\mathcal{J} U p\|_{\mathrm{F}}^2. \tag{14}$$

Apply the forward feature map and its strict adjoint using the same cached accepted-point factors. The norm and action then describe one computational graph, instead of separate expressions related by projector identities such as $P^{\mathsf{T}} SP = SP$ that need not hold to working precision for computed factors. No reduced metric matrix, inverse, eigendecomposition, or coordinate whitening is introduced by Eq. (14). Its application has shared $O(N_{\mathrm{BF}}(n_i + n_a))$ tangent/cotangent storage and leading work $O(N_{\mathrm{BF}}^2(n_i + n_a) + N_{\mathrm{BF}}(n_i + n_a)^2)$. The existing accepted-point AO and small orbital factors are reused; no large array is replicated per OpenMP worker. Matrix products are explicitly grouped to apply each AO operator to the narrow tangent or cotangent first: the implementation never forms LQ_a or $Q_a^{\mathsf{T}} L^{\mathsf{T}}$ during an action. Such products would introduce avoidable $O(N_{\mathrm{BF}}^3)$ work despite the factorized mathematical formula.

Literal bitwise symmetry is not asserted. This change removes a distinct forward/adjoint graph mismatch, without changing the AH symmetry certificate, orbital coordinates, Hessian–vector product, or convergence tolerances. The independent diagnostic constructs the feature Gram matrix at a captured molecular point; that dense matrix is a diagnostic only, never a production optimizer representation.

Operation	Implementation
Forward map	`physical_feature()`
Strict adjoint	`feature_adjoint()`
Metric composition	`apply()`

The previous expanded action and its unused caches are deleted.